Problem Statement :-

PROBLEM :- Design a synchronous counter that has the following sequence

000 , 010 , 101 , 110 and repeat

Step 1 :-

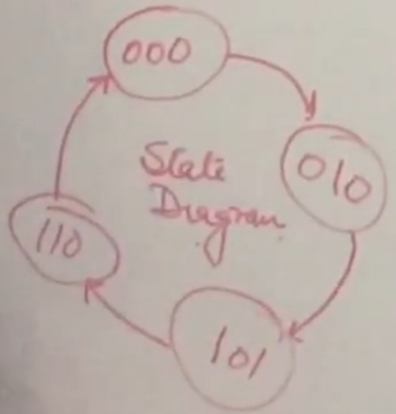
No of FF = No. of Bits

3 FFs , D-Type FF

Step 2 : Ext. D FF

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

Step 3 : State Diagram



Excitation table & K-map for ckt :-

PS			NS					
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0	D_2	D_1	D_0
0	0	0	0	1	0	0	1	0
0	0	1	X	X	X	X	X	X
0	1	0	1	0	1	1	0	1
0	1	1	X	X	X	X	X	X
1	0	0	X	X	X	X	X	X
1	0	1	1	1	0	X	X	X
1	1	0	0	0	0	1	1	0
1	1	1	X	X	X	0	0	0
						X	X	X

$D_2 = Q_0 + \overline{Q}_2 Q_1$

Q_0 \ Q_2Q_1	00	01	11	10
0	0	1	0	X
1	X	X	X	1

$D_1 = Q_0 + \overline{Q}_1$

Q_0 \ Q_2Q_1	00	01	11	10
0	1	0	0	X
1	X	X	X	1

$D_0 = \overline{Q}_2 Q_1$

Q_0 \ Q_2Q_1	00	01	11	10
0	0	1	0	X
1	X	X	X	0

Final ckt for given problem :-